SEED ROUND

FarmSense

The Deterministic Farming Operating System

$100K	$2M	5%
TARGET INVESTMENT	PRE-MONEY VALUATION	MAX EQUITY

01 / THE PROBLEM

$500/AF Water Crisis · 89,000 AF Annual Depletion

The Hydro-Economic Crisis

- → San Luis Valley: 300,000 acres of irrigated agriculture
- → 89,000 acre-feet annual aquifer depletion
- → $500/AF groundwater pumping fee = 4x previous costs
- → Reservoir storage: 26% of historical capacity
- → 126-acre pivot: $25,200/year savings at 20% reduction

Legal Imperative

- → Water Court requires empirical proof, not estimates
- → June 29, 2026 trial: Subdistrict 1 compliance
- → Current telemetry insufficient for legal defense
- → Farmers face existential threat without data
- → Digital Water Ledger = court-admissible evidence

Regulatory Pressure

- → 1938 Rio Grande Compact enforcement
- → RGWCD "presumed compliant" status opportunity
- → Mandatory reporting: Automated compliance
- → Water rights trading: AllianceChain ledger
- → Compact compliance: Federal priority

02 / THE SOLUTION

Deterministic OS · Digital Water Ledger · Court-Grade Evidence

Technical Architecture

- → 11 domain-specific engines (explainable, not AI)
- → SPAC modeling: Soil-Plant-Atmosphere Continuum
- → MAD framework: Management Allowable Depletion
- → RTK GNSS: Sub-centimeter spatial accuracy
- → AES-256 encryption: Tamper-evident chain of custody

Edge-Resilient Mesh

- → Decentralized monolithic grid (no cloud dependency)
- → 900MHz CSS LoRa: 10km range, LPI/LPD emissions
- → 30-day "Black Box" SSD audit cache
- → Reflex Logic: Autonomous field decisions
- → Contested environment: Off-grid operation

Legal Defensibility

- → Blockchain ledger: AllianceChain PBFT consensus
- → Multi-sig: Farmer + RGWCD + automated attestation
- → 40-year retention: Matching hardware lifespan
- → Expert witness: Exportable evidence packages
- → Immutable timestamps: Prior appropriation encoded

03 / FINANCIAL ARCHITECTURE

Grants Fund Scale · Investment Funds Pilot Validation

Investment Structure

$25K minimum to start pilot (pro-rated equity)
$100K target = up to 5% equity
$2M pre-money valuation
Funds: CSU SLV RC pilot operations only

Grant Pipeline ($4M+ Target)

- Federal ESG: $500K-$2M (March 26 deadline)
- Gates Foundation: $1M-$5M (post-pilot)
- World Food Prize: $500K (May 1 nomination)
- Earthshot Prize: £1M (Q2 2026)
- NSF SBIR: $275K (Q2 2026)
- CWCB/BoR: $800K-$4.3M (water resilience)

🔴 **Critical: Federal ESG Deadline**

March 26, 2026 (19 days) — $500K-$2M DoD dual-use grant. Pilot success required for full drawdown.

15x
LEVERAGE ($100K → $1.5M+ GRANTS)

$810M
HARDWARE CAPEX AT 5% GLOBAL

$1.11B
ANNUAL ARR AT SCALE

Farmer Onboarding Funnel · 30:1 LTV:CAC Ratio

$1,200	$35,928	30:1	1.35%
TARGET CAC PER FIELD	6-YEAR LTV (ENTERPRISE)	LTV:CAC RATIO	AWARENESS → ENTERPRISE

FUNNEL STAGE	CONVERSION	PRICE	KEY TACTIC
Awareness → Interest	15%	—	Water Court trial news, CSU field days
Interest → Trial	60%	Free	2-field demo, 30-day test, PMT install
Trial → Basic	60%	$99/mo	Weekly savings alerts, ROI calculator
Basic → Enterprise	25%	$499/mo	Resolution Pop, water court pressure

Channel Strategy

Direct (60%): Founder-led demos, field technicians, self-serve web
Partnerships (40%): CSU SLV RC (5% rev share), RGWCD, pivot dealers (10%), ag insurers

05 / REVENUE MODEL

The "Resolution Pop" · SaaS + Trading Fees

TIER	RESOLUTION	PRICE	TARGET	CONVERSION TRIGGER
Free	50m	$0	Government baseline	Compliance hook
Basic	20m	$99/mo	Small farmers	Weekly savings alerts
Pro	10m	$199/mo	Commercial ag	VRI actuation, daily reports
Enterprise	1m	$499/mo	Water Court / ESG	Resolution Pop blur effect

Trading Revenue (AllianceChain)

- → PBFT blockchain: Peer-to-peer water rights
- → $25 + 0.5% per transaction fee
- → 256 projected trades/year (Subdistrict 1)
- → $12,800 annual trading revenue
- → Lock-in effect: Lifetime SaaS subscribers

Expansion Revenue

- → Additional pivot: +$499/mo
- → Drone audit (Resolution Pop): +$150/flight
- → Insurance verification: +$500/report
- → CSA (Corner-Swing): +$998/mo
- → Water trading: +$50/transaction

06 / MARKET OPPORTUNITY

Subdistrict 1 → Colorado → 5% Global Cropland

1,280
SUBDISTRICT 1 PIVOTS

$7.6M
SD1 ARR AT SCALE

15M
HECTARES (5% GLOBAL)

The 5% Global Target

300M hectares global irrigated cropland × 5% = 15M hectares (37M acres). At 126 acres per pivot = 294,167 pivots. Hardware CAPEX: $810M. Annual ARR at scale: $1.11B. Gates Foundation subsidizes smallholder deployment; Enterprise tiers cross-subsidize free baseline.

GEOGRAPHIC TIER	TIMELINE	PIVOTS	REVENUE
Subdistrict 1 (Pilot)	2026	2 → 1,280	$7.6M ARR
Colorado River Compact States	2027-2028	50,000	$300M ARR
High Plains Aquifer	2028-2029	200,000	$1.2B ARR
5% Global Irrigated	2030+	294,167	$1.11B ARR

07 / HARDWARE ASSESSMENT

5% Global Scale = 294K Pivots · $810M CAPEX

COMPONENT	UNITS (5%)	UNIT COST	TOTAL CAPEX	FUNCTION
PMT (Pivot Motion Tracker)	294,167	$1,112	$327M	Field hub, edge-EBK, GNSS
LRZ (Lateral Root-Zone Scout)	2,471,000	$54	$134M	Soil moisture mapping
PFA (Pressure & Flow Anchor)	294,167	$850	$250M	Wellhead telemetry
VFA (Vertical Field Anchor)	294,167	$158	$47M	48" deep-profile truth
DHU (District Hub)	2,942	$12,500	$37M	Regional mesh manager
RSS (Regional Superstation)	147	$98,000	$14M	Territory master, cloud
TOTAL HARDWARE CAPEX			**$810M**	—

Funding Structure at Scale

- **DoD/Federal ESG:** $800M+ (base resilience, dual-use validation, contested environment)
- **World Bank/Gates Foundation:** $200M+ (trans-boundary treaties, smallholder adaptation)
- **Enterprise SaaS:** Cross-subsidizes free 50m tier for global baseline coverage

08 / TRACTION

Pilot Deployed · Water Court Trial June 29 · 29 Patents

Hardware Stack (CSU Pilot)

- → 2 PMTs (Pivot Motion Tracker) — deployed
- → 2 PFAs (Pressure & Flow Anchor) — deployed
- → 2 VFAs (Vertical Field Anchor) — deployed
- → 16-20 LRZs (Lateral Root-Zone Scout) — deployed
- → Validation: <5% Kriging error target

IP Portfolio

- → 29+ patents filed (provisional)
- → Sensor fusion algorithms
- → Deterministic crop optimization
- → Real-time water usage systems
- → Edge computing nodes for ag IoT

Partnerships

- → CSU SLV Research Center — academic validation
- → RGWCD — regulatory pathway
- → Colorado School of Mines — hydrology
- → DoD ESTCP — dual-use funding
- → Gates Foundation (pipeline) — global scale

Identified Risks · Contingency Plans · Legal Protection

RISK	SEVERITY	MITIGATION	STATUS
VFA Telemetry Failure (900MHz→2.4GHz)	High (15)	DHU BOM includes CSS LoRa gateway; RF site survey	✓ Rectified
PMT GNSS Degradation (RTK loss)	Med (12)	Dual-RTK base stations; PPP fallback	⚠ Pre-pilot setup
PFA Flow Sensor Drift	Med (10)	Pre-install calibration; quarterly recalibration	⚠ Scheduled
DHU "Black Box" SSD Failure	Med (10)	RAID-1 mirror; weekly RSS backup	✓ Designed
Chain of Custody Breach	Med (10)	AES-256; tamper seals; dual-authorization	✓ Implemented
Farmer Adoption Resistance	Med (12)	CSU credibility; free trials; ROI guarantee	⚠ Monitor
Non-Dilutive Funding Collapse	Med (12)	Parallel 5+ grants; bridge financing; VC pivot	⚠ Track weekly

Legal Protection

- Formal MOU with CSU SLV RC (12-month notice)
- Comprehensive insurance: $150K equipment coverage
- Water rights attorney on retainer
- Annual third-party security audit (SOC 2 Type II equivalent)

Leadership · Colorado-Based · Deep Domain Expertise

Jeremy B. Beebe

Founder, CTO, Board Member

- → Building the future of agriculture through AI and robotics
- → 29+ patent portfolio architect
- → Deterministic systems specialist
- → San Luis Valley native knowledge

Johnathon Montez

CEO, Board Member

- → Strategic operations and business development
- → Grant strategy and federal relations
- → Partnership development (CSU, RGWCD, DoD)
- → Fundraising and investor relations

Fabian Gomez

CFO, Board Member

- → Financial operations and capital efficiency
- → Grant compliance and reporting
- → Hardware procurement and BOM optimization
- → Strategic financial planning

$25K Minimum · $100K Target · Pilot Operations Only

Pilot Budget: CSU San Luis Valley Research Center

CSU SLV RC Operations (personnel, logistics)	$45K (45%)
Academic Partnership & Data Validation	$25K (25%)
Legal Prep & Water Court Evidence	$20K (20%)
Contingency	$10K (10%)

Grant-Covered (Not Investment)

Hardware (2 PMTs, 2 PFAs, 2 VFAs, 16-20 LRZs = $12.5K), software development, R&D, patents — all covered by Federal ESG, NSF SBIR, CWCB grants.

Milestones Unlocked

- → **April 2026:** Pilot deploy (2 fields)
- → **June 2026:** Validation (<5% Kriging error)
- → **June 29:** Water Court trial
- → **July 2026:** Federal ESG drawdown
- → **Q3 2026:** Gates LOI submission

$1.5M+

Grant runway unlocked by pilot success

$25K Minimum · $100K Target

Fund the validation event that unlocks $1.5M+ in grants.
Grants cover $810M hardware scale. Investors capture 5% of $1.11B ARR potential.

CONTACT
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TRIAL DATE
June 29, 2026

FEDERAL ESG DEADLINE
March 26, 2026